

Commutative Algebra & Algebraic Geometry

Exercise 5

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5.1 Problem

- (1) Let $a, b \in \mathbb{C}$ be nonzero, and L be the line in $\mathbb{C}^2$ defined by $ax + by = 0$. Let $P: \mathbb{C}^2 \rightarrow \mathbb{C}^2$ be orthogonal projection onto L , with respect to the standard inner product.
 - (i) Find a morphism of $\mathbb{C}$ -algebras $\varphi: \mathbb{C}[z, w] \rightarrow \mathbb{C}[x, y]$ which induces a morphism on closed points $|\varphi^*|: |\mathbb{A}_{\mathbb{C}}^2| \rightarrow |\mathbb{A}_{\mathbb{C}}^2|$ equal to P , identifying $|\mathbb{A}_{\mathbb{C}}^2|$ with $\mathbb{C}^2$.
 - (ii) Let $\pi: \mathbb{C}[x, y] \rightarrow \mathbb{C}[x, y]/(xy - 1)$ be the projection. Show that if $a, b \neq 0$, then $\pi \circ \varphi$ is finite.
- (2) Let $R = \mathbb{C}[x, y]/(y^2z - x^3 - 1)$.
 - (i) Show that R is not a finite $\mathbb{C}[x, y]$ -algebra.
 - (ii) Find $x', y' \in R$ such that R is a finite $\mathbb{C}[x', y']$ -algebra.

(1)

- (i) Note that L is spanned by $(b, -a)$ and thus P is given by

$$P(\lambda, \mu) = \frac{(\lambda, \mu) \cdot (b, -a)}{|(b, -a)|^2} (b, -a) = \frac{b\lambda - a\mu}{a^2 + b^2} (b, -a).$$

And so we want a φ such that $\varphi^* = \varphi^{-1}$ maps

$$(x - \lambda, y - \mu) \mapsto \left(z - \frac{b^2\lambda - ab\mu}{a^2 + b^2}, w - \frac{a^2\mu - ab\lambda}{a^2 + b^2} \right) = (z - P_1, w - P_2).$$

Since we are dealing with maximal ideals, $\varphi^{-1}(x - \lambda, y - \mu) = (z - P_1, w - P_2)$ iff $\varphi(z - P_1, w - P_2) \subseteq (x - \lambda, y - \mu)$ since then $\varphi^{-1}(x - \lambda, y - \mu) \supseteq (z - P_1, w - P_2)$ and since φ^{-1} maps maximal ideals to maximal ideals, we must have equality.

Since $\mathbb{C}[z, w]$ is the free $\mathbb{C}$ -algebra on two generators, we need only define φ on z, w . We will try φ which maps $\varphi z = \alpha x + \beta y$, $\varphi w = \gamma x + \delta y$. Then

$$\varphi\left(z - \frac{b^2\lambda - ab\mu}{a^2 + b^2}\right) = \alpha x + \beta y - \frac{b^2\lambda - ab\mu}{a^2 + b^2}.$$

We want this to be in $(x - \lambda, y - \mu)$, so let us write this as

$$= \alpha\left(x - \lambda \cdot \frac{1}{\alpha} \cdot \frac{b^2}{a^2 + b^2}\right) + \beta\left(y - \mu \cdot \frac{1}{\beta} \cdot \frac{-ab}{a^2 + b^2}\right).$$

So we see that

$$\alpha = \frac{b^2}{a^2 + b^2}, \quad \beta = -\frac{ab}{a^2 + b^2}$$

do the trick. Similarly

$$\gamma = -\frac{ab}{a^2 + b^2}, \quad \delta = \frac{a^2}{a^2 + b^2}.$$

These are not dependent on λ, μ and thus φ is well-defined and satisfies the desired property. Notice that if we only assume that not both a, b are zero, then these definitions work as well (I assume this is what the question meant to ask, but “both not zero” and “not both zero” mean different things).

(ii) Notice that

$$z \cdot x = (\alpha x + \beta y)x = \alpha x^2 + \beta xy = \alpha x^2 + \beta.$$

Since $b \neq 0$, $\alpha \neq 0$ and thus is invertible (since it is a complex number). Thus

$$t^2 - \frac{z}{\alpha}t + \frac{\beta}{\alpha} \in \mathbb{C}[z, w][t]$$

is a monic polynomial with root x , so x is integral over $\mathbb{C}[z, w]$. Similarly

$$w \cdot y = (\gamma x + \delta y)y = \delta y^2 + \gamma,$$

and thus $t^2 - w/\delta t + \gamma/\delta$ is a monic polynomial over $\mathbb{C}[z, w]$ with root y , so y is integral over $\mathbb{C}[z, w]$.

Furthermore, x, y generate $\mathbb{C}[x, y]/(xy - 1)$ so it is integral over $\mathbb{C}[z, w]$ and fg, so it is finite as desired.

(2)

- (i) Note that we can embed R into $\text{Frac}(\mathbb{C}[x, y])$ by mapping z to $\frac{x^3+1}{y} \notin \mathbb{C}[x, y]$. Identifying R with its image in the field of fractions, $\mathbb{C}[x, y] \subsetneq R \subseteq \text{Frac}(\mathbb{C}[x, y])$. Since $\mathbb{C}$ is a field, $\mathbb{C}[x, y]$ is a UFD and thus normal, so it is integrally closed in its field of fractions. Thus R cannot be integral over $\mathbb{C}[x, y]$, since it is a proper extension of it. And so it cannot be finite over $\mathbb{C}[x, y]$.
- (ii) We recall the proof of Noether's normalization lemma. Consider $f = y^2z - x^3 - 1$; its multi-degrees (of nonzero coefficients) are $(3, 0, 0)$, $(0, 2, 1)$, and $(0, 0, 0)$. Then we set $k_3 = 1$,

$$k_2 = 1 + \max \{k_3 \cdot 0, k_3 \cdot 0, k_3 \cdot 1\} = 2,$$

$$k_1 = 1 + \max \{k_2 \cdot 0 + k_3 \cdot 0, k_2 \cdot 0 + k_2 \cdot 2, k_2 \cdot 0 + k_3 \cdot 0\} = 5.$$

Then

$$x' = x - z^{k_1} = x - z^5, \quad y' = y - z^{k_2} = y - z^2$$

does the trick.

5.2 Problem

Let k be a field.

- (1) Show that k is algebraically closed iff for every $n \in \mathbb{N}$ and nonconstant $f \in k[x_1, \dots, x_n]$, f has a zero.
- (2) Let A be a fg k -algebra and $\mathfrak{m} \leq A$ a maximal ideal. We assume the third formulation of Nullstellensatz (polynomial rings over k are Jacobson), and wish to prove the first ($A/\mathfrak{m}$ is a finite k -algebra).
 - (i) Show that A satisfies the third form of Nullstellensatz. That is, it is Jacobson: $\mathfrak{p} = \text{Jac}_A(\mathfrak{p})$ for all prime ideals $\mathfrak{p} \leq A$.
 - (ii) Show that if $B \subseteq A$ is a fg k -subalgebra, then $\mathfrak{m} \cap B$ is a maximal ideal of B .

(iii) Suppose that $A/\mathfrak{m}$ is not a finite extension of k . Find an element $x \in A$ such that x is transcendental in $A/\mathfrak{m}$. Conclude that $\mathfrak{m} \cap k[x] = 0$ and reach a contradiction.

(1) If every non-constant multivariate polynomial has a zero, then every non-constant single-variable polynomial has a zero, so k is algebraically closed. Conversely if $f \in k[\bar{x}]$ is nonconstant, it is not a unit, and thus $(f) \neq (1)$. By Nullstellensatz, $Z(f) = Z((f)) \neq \emptyset$, so f has a zero.

(2)

(i) Since A is fg, there is a polynomial k -algebra $k[x_1, \dots, x_n]$ as well as a surjective k -algebra morphism $k[x_1, \dots, x_n] \rightarrow A$, so A is isomorphic to $k[\bar{x}]/I$ for some ideal I . The third form of Nullstellensatz says that $k[\bar{x}]$ is Jacobson, we claim that the quotient of a Jacobson ring is Jacobson, in particular A is.

If B is a Jacobson ring and $I \leq B$ is an ideal, let $\mathfrak{p} \in B/I$ be prime. Let $\pi: B \rightarrow B/I$ be the projection so by correspondence $\mathfrak{p} = \pi(\tilde{\mathfrak{p}})$ for a prime ideal $I \leq \tilde{\mathfrak{p}} \leq B$. Then

$$\text{Jac}_{B/I}(\mathfrak{p}) = \bigcap_{\mathfrak{p} \leq \mathfrak{m}} \mathfrak{m},$$

where the intersection ranges over all maximal ideals containing $\mathfrak{p}$. By correspondence the maximal ideals in B/I correspond to maximal ideals in B containing I , and so the maximal ideals containing $\mathfrak{p}$ correspond to the maximal ideals containing $\tilde{\mathfrak{p}}$:

$$\text{Jac}_{B/I}(\mathfrak{p}) = \bigcap_{\tilde{\mathfrak{p}} \leq \tilde{\mathfrak{m}}} \pi(\tilde{\mathfrak{m}}) = \pi \left(\bigcap_{\tilde{\mathfrak{p}} \leq \tilde{\mathfrak{m}}} \tilde{\mathfrak{m}} \right) = \pi(\text{Jac}_B(\tilde{\mathfrak{p}})),$$

and this is equal to $\pi(\tilde{\mathfrak{p}}) = \mathfrak{p}$ since B is Jacobson. So B/I is indeed Jacobson, as desired.

(ii) This was listed as optional, so I unfortunately decided not to do it, I apologize!

(iii) $A/\mathfrak{m}$ is a field and a fg k -algebra, suppose $A/\mathfrak{m}$ is not a finite extension of k . Suppose $x_1, \dots, x_n \in A$ generate A and thus their projections generate $A/\mathfrak{m}$. If these are all algebraic over k , then $A/\mathfrak{m}$ must be finite (if $x_1, \dots, x_n$ are algebraic over k then $k[x_1, \dots, x_n]$ is a finite extension of k), so one must be transcendental, let it be a .

By the above point since $k[a] \subseteq A$ is a fg subalgebra, $k[a] \cap \mathfrak{m}$ is a maximal ideal of $k[a]$. But since $a \in A$ is transcendental in $A/\mathfrak{m}$, there can be no single-variable polynomials $p \in k[x]$ such that $p(a) = 0$ in $A/\mathfrak{m}$. That is, $p(a) \notin \mathfrak{m}$ for all non-zero $p \in k[x]$. Thus $k[a] \cap \mathfrak{m} = 0$ is a maximal ideal of $k[a]$.

On the other hand since a is transcendental, $k[a]$ is isomorphic to the $k[x]$ as k -algebras. 0 is not a maximal ideal of $k[x]$, and thus not a maximal ideal of $k[a]$, a contradiction.

5.3 Problem

Let R be a ring.

- (1) Let M, N be free R -modules with respective bases $\{a_i\}_{i \in I}, \{b_j\}_{j \in J}$. Show that $M \otimes_R N$ is a free R -module with basis $\{a_i \otimes b_j\}_{i \in I, j \in J}$.
- (2) Let A be an R -algebra, show that $A \otimes_R (R[x]) \cong A[x]$.

- (3) Let $\varphi: R \rightarrow A$ be a morphism of rings and $I \leq R$ an ideal. Show that $A \otimes_R (R/I) \cong A/IA$.
- (4) Assume R is a PID, and let $a, b \in R$. Compute $(R/(a)) \otimes_R (R/(b))$, and explain what this means for $R = \mathbb{Z}$.
- (5) Let L/K be a finite Galois field extension. Show that there is an isomorphism of K -algebras

$$L \otimes_R L \rightarrow \prod_{\sigma \in \text{Gal}(L/K)} L, \quad a \otimes b \mapsto (a\sigma(b))_{\sigma \in \text{Gal}(L/K)}.$$

- (6) Let A be an R -algebra and $S \subseteq R$ a multiplicative subset. Let M be an R -module, show that $(S^{-1}A) \otimes_R M \cong S^{-1}(A \otimes_R M)$.

- (1) Let P be the free R -module generated by $\{a_i \otimes b_j\}_{i \in I, j \in J}$. Define $M \times N \rightarrow P$ by $(a_i, b_j) \mapsto a_i \otimes b_j$, and extending by linearity. This is well-defined since a_i, b_j form bases. This is R -bilinear, since the tensor product is, and thus forms an R -algebra morphism $M \otimes_R N \rightarrow P$.

Then define $P \rightarrow M \otimes_R N$ by mapping $a_i \otimes b_j$ to itself (the former viewed as a basis element of P , the latter as an element of the tensor product).

These are clearly inverses (since $a_i \otimes b_j$ generates the tensor product and forms a basis for P , it is sufficient to check this just for elements of this form), and thus $M \otimes_R N$ is free with basis $a_i \otimes b_j$ (following the basis from P).

- (2) Define a map $A \times R[x]$ which maps $(a, p(x))$ to $ap(x) \in A[x]$, where

$$a \sum_k b_k x^k = \sum_k ab_k x^k \in A[x],$$

(since $b_k \in R$, and A is an R -algebra, $ab_k \in A$ so this is indeed a polynomial over A .) This is R -bilinear clearly and thus gives us a map $\Phi: A \otimes_R R[x] \rightarrow A[x]$. It is a ring morphism since $(ap)(bq) = ab(pq)$, and thus gives a morphism of R -algebras.

In the other direction define $\Psi: A[x] \rightarrow A \otimes_R R[x]$ by

$$\Psi \sum_k a_k x^k = \sum_k (a_k \otimes x^k).$$

This is an R -algebra morphism as well: it is R -linear since $ra_k x^k$ is mapped to $ra_k \otimes x^k = a_k \otimes rx^k$. And it is a ring morphism since

$$\Psi \sum_k \sum_i a_i b_{k-i} x^k \mapsto \sum_k \sum_i a_i b_{k-i} \otimes x^k = \sum_k \sum_i (a_i \otimes x^i)(b_{k-i} \otimes x^{k-i}) = \left(\sum_k a_k \otimes x^k \right) \left(\sum_k b_k \otimes x^k \right)$$

These are inverses, since

$$\Phi \Psi a_k x^k = \Phi(a_k \otimes x^k) = a_k x^k,$$

and

$$\Psi \Phi a \otimes x^k = \Psi ax^k = a \otimes x^k.$$

Thus we have the desired isomorphisms.

- (3) Let us define an R -algebra morphism $A \rightarrow A \otimes_R (R/I)$ by $a \mapsto a \otimes 1$. Then for $x \in I$, we have that $\varphi(x) \mapsto \varphi(x) \otimes 1 = 1 \otimes x$ (since $\varphi(x) = x \cdot 1$), which is zero as $x \in R/I$ is zero. Thus IA lies in the kernel of this map, so we can quotient it to give an R -algebra morphism $\Phi: A/IA \rightarrow A \otimes_R (R/I)$.

Define $A \times (R/I) \rightarrow A/IA$ by $(a, r+I) \mapsto (ar) + IA$. This is well-defined since for $x \in I$, $ax \in IA$. Furthermore it is R -bilinear clearly and thus defines an R -module morphism $\Psi: A \otimes_R (R/I) \rightarrow A/IA$. It is also a ring morphism, but this is immediate as we will show it is Φ 's inverse.

Indeed,

$$\Phi \Psi a \otimes (r + I) = \Phi(ar + IA) = ar \otimes 1 = a \otimes (r + I),$$

and

$$\Psi \Phi(a + IA) = \Psi(a \otimes 1) = a + IA,$$

as desired.

- (4) We claim that in general, for ideals $I, J \leq R$,

$$R/I \otimes R/J \cong R/(I + J).$$

By the above point,

$$R/I \otimes R/J \cong \frac{R/I}{J \cdot R/I}.$$

And $J \cdot R/I = (I + J)/I$, since for $j \in J$ and $x \in R$, $j(x + I) = jx + I$ where $jx \in J$ and thus $jx + I \in (I + J)/I$. Conversely for $i + j + R \in (I + J)/I$, $i + j + I = j + I = j(1 + I) \in J \cdot (R/I)$. Thus by the isomorphism theorems, there is an isomorphism of R -algebras

$$R/I \otimes R/J \cong \frac{R/I}{J \cdot R/I} = \frac{R/I}{(I + J)/I} \cong R/(I + J).$$

In our case, $(a) + (b) = (c)$ where c is a gcd of a and b and so

$$R/(a) \otimes R/(b) \cong R/(c).$$

When $R = \mathbb{Z}$ we literally have

$$\mathbb{Z}/n \otimes_{\mathbb{Z}} \mathbb{Z}/m \cong \mathbb{Z}/\gcd(n, m).$$

- (5) Let $G = \text{Gal}(L/K)$. Since L/K is finite Galois, it is generated by a single element $L = K[a]$. Let p_a be a 's minimal polynomial, which splits since the extension is Galois: $p_a(x) = \prod_{\sigma \in G} (x - \sigma(a))$. Thus

$$L \otimes_K L = L \otimes_K K[a] \cong L \otimes_K \frac{K[x]}{(p_a)}.$$

Now, by the above points we have

$$A \otimes_R \frac{R[x]}{I} \cong A \otimes_R R[x] \otimes_{R[x]} \frac{R[x]}{I} \cong A[x] \otimes_{R[x]} \frac{R[x]}{I} \cong \frac{A[x]}{I \cdot A[x]}.$$

Chasing definitions, the isomorphism is given by $a \otimes (p(x) + I) \mapsto ap(x) + IA[x]$. In our case, $(p_a) \cdot L[x] = (p_a)$ where the right-hand ideal is in $L[x]$, and thus

$$L \otimes_K L \cong \frac{L[x]}{(p_a)}.$$

Recall that p_a splits over L : $p_a = \prod_{\sigma} (x - \sigma(a))$. The $\sigma(a)$'s are all distinct (since p_a splits), and thus the ideals $(x - \sigma(a))$ are coprime (as they are distinct maximal ideals), and thus by the Chinese remainder theorem:

$$L \otimes_K L \cong \frac{L[x]}{(\prod_{\sigma} (x - \sigma(a)))} \cong \prod_{\sigma} \frac{L[x]}{(x - \sigma(a))} \cong \prod_{\sigma} L.$$

So we have a composition of isomorphisms,

$$L \otimes_K L \xrightarrow{(1)} L \otimes_K \frac{K[x]}{(p_a)} \xrightarrow{(2)} \frac{L[x]}{(p_a)} \xrightarrow{(3)} \prod_{\sigma} \frac{L[x]}{(x - \sigma)} \xrightarrow{(4)} \prod_{\sigma} L.$$

For $\beta \in L = K[a]$, let $p_{\beta}(x) \in K[x]$ be a polynomial such that $p_{\beta}(a) = \beta$. Chasing definitions, we see

$$\alpha \otimes \beta \xrightarrow{(1)} \alpha \otimes (p_{\beta}(x) + (p_a)) \xrightarrow{(2)} \alpha p_{\beta}(x) + (p_a) \xrightarrow{(3)} (\alpha p_{\beta}(x) + (x - \sigma(a)))_{\sigma} \xrightarrow{(4)} (\alpha p_{\beta}(\sigma a))_{\sigma}.$$

For $\sigma \in G$ and $p \in K[x]$ we have $p(\sigma(x)) = \sigma(p(x))$, and thus $\alpha p_{\beta}(\sigma a) = \alpha \sigma(p_{\beta}(a)) = \alpha \sigma(\beta)$. So our isomorphism is given by

$$\alpha \otimes \beta \mapsto (\alpha \sigma(\beta))_{\sigma},$$

as desired.

(6) We have that $S^{-1}A \cong S^{-1}R \otimes_R A$ as R -algebras. Thus

$$(S^{-1}A) \otimes_R M \cong (S^{-1}R \otimes_R A) \otimes_R M \cong S^{-1}R \otimes_R (A \otimes_R M) \cong S^{-1}(A \otimes_R M).$$

This is true also when we take $S \subseteq A$ multiplicative (which may have been the intended question?). Indeed,

$$S^{-1}A \otimes_R M \cong (S^{-1}A \otimes_A A) \otimes_R M \cong S^{-1}A \otimes_A (A \otimes_R M) \cong S^{-1}(A \otimes_R M).$$

The last isomorphism is due to the natural isomorphism $S^{-1}\bullet \cong S^{-1}A \otimes_A \bullet$.

5.4 Problem

Let R be a ring, $S = \text{Spec}(R)$, and A and B be R -algebras with $X = \text{Spec}(A)$, $Y = \text{Spec}(B)$.

- (1) Assume $R \rightarrow A, R \rightarrow B$ are surjective, so that $X \rightarrow S, Y \rightarrow S$ are closed embeddings.
 - (i) Show that $X \times_S Y$ is homeomorphic to $X \cap Y$.
 - (ii) Let $R = \mathbb{C}[x, y]$, $A = \mathbb{C}[x, y]/(y - f)$ for $f \in \mathbb{C}[x]$, and $B = \mathbb{C}[x, y]/(y)$. Prove $A \otimes_R B$ is a complex vector space whose dimension equals the number of zeroes of f , with multiplicity.
- (2) Let $\mathfrak{p} \in S$ and suppose $A = k(\mathfrak{p})$. Let φ be the morphism $Y \rightarrow S$, and show that $X \times_S Y$ is homeomorphic to $\varphi^{-1}\mathfrak{p}$.
- (3) Assume $R = \mathbb{C}[x]$, $A = \mathbb{C}[y]$, and the morphism $R \rightarrow A$ is $x \mapsto y^2$. Prove that $X \times_S X$ has two irreducible components.
- (4) By the definition of a pullback, the identity $X \rightarrow X$ yields a morphism $X \rightarrow X \times_S X$ called the **diagonal** and denoted Δ . Show that Δ is a closed embedding.

(1)

- (i) The closed embeddings $X, Y \rightarrow S$ yield homeomorphisms $X \cong V_R(I)$ and $Y \cong V_R(J)$ for ideals I, J (the kernels of the maps from $R \rightarrow A, B$). Then $X \cap Y$ is $V_R(I) \cap V_R(J) = V_R(I + J)$, which is an affine scheme as $V_R(I + J) \cong \text{Spec}(R/(I + J))$. So we have $A \cong R/I, B \cong R/J$, and then as we showed

$$A \otimes_R B \cong (R/I) \otimes_R (R/J) \cong R/(I + J),$$

which yields a homeomorphism

$$X \times_S Y = \operatorname{Spec}(A \otimes_R B) \cong \operatorname{Spec}(R/(I + J)) \cong V_R(I + J) = V_R(I) \cap V_R(J) = X \cap Y$$

as desired.

(ii) We have by above

$$A \otimes_R B = \frac{\mathbb{C}[x, y]}{(y - f)} \otimes_{\mathbb{C}[x, y]} \frac{\mathbb{C}[x, y]}{(y)} \cong \frac{\mathbb{C}[x, y]}{(y, y - f)} \cong \mathbb{C}[x]/(f).$$

Note that as $\mathbb{C}$ -vector spaces

$$\mathbb{C}[x]/(f) \cong \{g \in \mathbb{C}[x] \mid \deg g < \deg f\},$$

and the right-hand side is a complex vector space of dimension $\deg f$. The number of f 's zeroes, counted with multiplicity, is just $\deg f$.

(2) From homework 4 question 3, we know that $\varphi^{-1}(\mathfrak{p})$ is homeomorphic to $\operatorname{Spec}(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}})$. The pullback is

$$X \times_S Y = \operatorname{Spec}(A \otimes_R B) = \operatorname{Spec}(k(\mathfrak{p}) \otimes_R B) = \operatorname{Spec}\left(\frac{R_{\mathfrak{p}}}{\mathfrak{p}R_{\mathfrak{p}}} \otimes_R B\right).$$

Now, from the previous question we know $A/IA \cong A \otimes_R R/I$ and $S^{-1}A \otimes_R B \cong S^{-1}(A \otimes_R B)$, so

$$\frac{R_{\mathfrak{p}}}{\mathfrak{p}R_{\mathfrak{p}}} \otimes_R B \cong (R/\mathfrak{p}) \otimes_R R_{\mathfrak{p}} \otimes_R B \cong (R/\mathfrak{p}) \otimes_R (R \otimes_R B)_{\mathfrak{p}} \cong (R/\mathfrak{p}) \otimes_R B_{\mathfrak{p}} \cong \frac{B_{\mathfrak{p}}}{\mathfrak{p}B_{\mathfrak{p}}}.$$

Thus this induces a homeomorphism

$$X \times_S Y = \operatorname{Spec}\left(\frac{R_{\mathfrak{p}}}{\mathfrak{p}R_{\mathfrak{p}}} \otimes_R B\right) \cong \operatorname{Spec}\left(\frac{B_{\mathfrak{p}}}{\mathfrak{p}B_{\mathfrak{p}}}\right) \cong \varphi^{-1}(\mathfrak{p}),$$

as desired.

(3) Recall that an irreducible component of $\operatorname{Spec}(A)$ is the vanishing set of a minimal prime ideal. So we must find the minimal prime ideals of $A \otimes_R A$:

$$A \otimes_R A \cong \mathbb{C}[y] \otimes_{\mathbb{C}[x]} \mathbb{C}[y] \cong \frac{\mathbb{C}[y_1, y_2]}{(y_1^2 - y_2^2)}.$$

The final equality is due to the tensor $\mathbb{C}[y_1] \otimes_{\mathbb{C}} \mathbb{C}[y_2] \cong \mathbb{C}[y_1, y_2]$ and having to identify the image of x in both cases. Explicitly, $\mathbb{C}[y_1] \times \mathbb{C}[y_2] \rightarrow \mathbb{C}[y_1, y_2]/(y_1^2 - y_2^2)$ given by $(f(y_1), g(y_2)) \mapsto f(y_1)g(y_2)$ is $\mathbb{C}[x]$ -bilinear since $(xf(y_1), g(y_2))$ is mapped to $y_1^2 f(y_1)g(y_2) = y_2^2 f(y_1)g(y_2)$, the image of $(f(y_1), xg(y_2))$. Thus it yields a map $\mathbb{C}[y_1] \otimes_{\mathbb{C}[x]} \mathbb{C}[y_2] \rightarrow \mathbb{C}[y_1, y_2]/(y_1^2 - y_2^2)$.

The obvious map $\mathbb{C}[y_1, y_2] \rightarrow \mathbb{C}[y_1] \otimes_{\mathbb{C}[x]} \mathbb{C}[y_2]$ mapping $y_1 \mapsto y_1 \otimes 1, y_2 \mapsto 1 \otimes y_2$ maps $y_1^2 - y_2^2$ to zero (since both are equal to x), and thus quotients to give a map $\mathbb{C}[y_1, y_2]/(y_1^2 - y_2^2)$ to the tensor product. These two maps are clearly inverses.

Thus we have

$$A \otimes_R A \cong \frac{\mathbb{C}[y_1, y_2]}{(y_1^2 - y_2^2)}.$$

Note that $y_1^2 - y_2^2 = (y_1 - y_2)(y_1 + y_2)$, a factorization of irreducibles. Let $\mathfrak{p}_1 = (y_1 - y_2)/(y_1^2 - y_2^2)$ and $\mathfrak{p}_2 = (y_1 + y_2)/(y_1^2 - y_2^2)$, so they are prime (as $y_1 \pm y_2$ are irreducible). Furthermore, $\mathfrak{p}_1 \mathfrak{p}_2 = 0$ and thus they must be the minimal prime ideals. This is because any other prime ideal contains 0 and thus by primality one of $\mathfrak{p}_1$ or $\mathfrak{p}_2$, and $\mathfrak{p}_1, \mathfrak{p}_2$ are distinct.

Thus the tensor has two distinct minimal prime ideals, and thus two irreducible components.

(4) We showed explicitly that given maps $\varphi, \psi: \operatorname{Spec}(C) \rightarrow \operatorname{Spec}(A), \operatorname{Spec}(B)$, then the map $\operatorname{Spec}(C) \rightarrow \operatorname{Spec}(A \otimes_R B)$ is induced by $A \otimes_R B \rightarrow C, a \otimes b \mapsto \varphi(a)\psi(b)$. In our case $C = A = B$ and $\varphi = \psi = \operatorname{id}$ so Δ is $(a \otimes b \mapsto ab)^*$. The map $a \otimes b \mapsto ab$ is clearly surjective, and thus Δ is a closed embedding as desired.

5.5 Problem

For rings R, S , an (R, S) -bimodule is an Abelian group which is both an R - and S -module, such that their operations commute. Denote the category of (R, S) -bimodules by $\text{Mod}_{R,S}$. Let $M \in \text{Mod}_R, K \in \text{Mod}_{R,S}, N \in \text{Mod}_S$.

- (1) Show that there is a natural S -module structure on $M \otimes_R K$, which gives us a functor $\bullet \otimes_R K: \text{Mod}_R \rightarrow \text{Mod}_S$.
- (2) Find a natural R -module structure on $\text{hom}_S(K, N)$, yielding a functor $\text{hom}_S(K, \bullet): \text{Mod}_S \rightarrow \text{Mod}_R$.
- (3) Show that there is an adjunction $\bullet \otimes_R K \vdash \text{hom}_S(K, \bullet)$.
- (4) Now assume S is an R -algebra (and thus also an (R, S) -bimodule). Denote $\bullet \otimes_R S$ by $\bullet^S$ and call it **extension by scalars**, $\bullet^S: \text{Mod}_R \rightarrow \text{Mod}_S$. Similarly we can view N as an R -module via **restriction**, yielding a functor $\bullet_R: \text{Mod}_S \rightarrow \text{Mod}_R$. Show that $\bullet_R$ is naturally isomorphic to $\text{hom}_S(S, \bullet)$, and thus there is an adjunction $\bullet^S \vdash \bullet_R$.

- (1) The natural structure is given by $s \cdot (m \otimes k) = m \otimes (sk)$ and extending by linearity. Assuming this is well-defined this gives an S -module structure, since

$$s_1 s_2 (m \otimes k) = m \otimes (s_1 s_2 k) = s_1 (m \otimes (s_2 k)) = s_1 (s_2 (m \otimes k)).$$

To show this is well-defined and linear, we note that the map $M \times K \rightarrow M \otimes K$ given by $s(m, k) = m \otimes (sk)$ is R -bilinear. This is clearly additive and bi-multiplicative since

$$s(rm, k) = (rm) \otimes (sk) = r(m \otimes (sk)) = rs(m, k) = m \otimes (rsk) = m \otimes (srk) = s(m, rk).$$

And it is functorial: if $f: M \rightarrow M'$ is an R -module morphism, then $f \otimes K: M \otimes K \rightarrow M' \otimes K$ is an S -module morphism since

$$f \otimes K(s(m \otimes k)) = f \otimes K(m \otimes sk) = f(m) \otimes sk = s(f(m) \otimes k) = s(f \otimes K)(m \otimes k),$$

and $\text{id} \otimes K$ is the identity.

- (2) The sum of two S -module morphisms is an S -module morphism clearly. And for $f \in \text{hom}_S(K, N)$ and $r \in R$ define rf by $(rf)(k) = f(rk)$. This is an S -module morphism since it is additive:

$$(rf)(k + k') = f(rk + rk') = f(rk) + f(rk') = (rf)(k) + (rf)(k'),$$

and multiplicative

$$(rf)(sk) = f(rsk) = f(sr k) = sf(rk) = s(rf)(k).$$

This turns $\text{hom}_S(K, N)$ into an R -module since $r(f + f')(k) = (f + f')(rk) = f(rk) + f'(rk) = (rf + rf')(k)$ so multiplication is additive. And

$$((rr')f)(k) = f(rr'k) = f(r'r k) = (r'f)(rk) = (r(r'f))(k) \implies (rr')f = r(r'f).$$

This is also functorial since if $F: N \rightarrow N'$ is an S -module morphism then $F^*: \text{hom}_S(K, N) \rightarrow \text{hom}_S(K, N')$ given by $f \mapsto F \circ f$ is an R -module morphism. Indeed,

$$F \circ (f + f') = F \circ f + F \circ f'$$

so it is additive, and

$$(F \circ (rf))(k) = F((rf)(k)) = F(f(rk)) = r(F \circ f)(k) \implies F^*(rf) = rF^*(f),$$

so it is multiplicative. Clearly id^* is the identity as well.

- (3) We show that $\text{hom}_S(M \otimes_R K, N) \cong \text{hom}_R(M, \text{hom}_S(K, N))$ naturally. For an S -map $F: M \otimes_R K \rightarrow N$, define $\hat{F}: M \rightarrow \text{hom}_S(K, N)$ by

$$\hat{F}(m)(k) = F(m \otimes k), \quad m \in M, k \in K.$$

This is well-defined: $\hat{F}(m)$ is an S -module morphism:

$$\hat{F}(m)(k + k') = F(m \otimes k + m \otimes k') = \hat{F}(m)(k) + \hat{F}(m)(k'),$$

and

$$\hat{F}(m)(sk) = F(m \otimes sk) = F(s(m \otimes k)) = sF(m \otimes k) = s\hat{F}(m)(k).$$

And conversely given $G: M \rightarrow \text{hom}_S(K, N)$, define $\tilde{G}: M \otimes_R K \rightarrow N$ by

$$\tilde{G}(m \otimes k) = G(m)(k).$$

This is well-defined since the map $M \times K \rightarrow N$ given by $G'(m, k) = G(m)(k)$ is R -bilinear. It is additive:

$$G'(m + m', k) = G(m + m')(k) = (G(m) + G(m'))(k) = G(m)(k) + G(m')(k) = G'(m, k) + G'(m', k),$$

and

$$G'(m, k + k') = G(m)(k + k') = G(m)(k) + G(m)(k') = G'(m, k) + G'(m, k').$$

And it is R -bilinear:

$$G'(rm, k) = G(rm)(k) = rG(m)(k) = G(m)(rk) = G'(m, rk).$$

And clearly $\tilde{\bullet}$ and $\hat{\bullet}$ are inverses:

$$\tilde{\hat{F}}(m \otimes k) = \hat{F}(m)(k) = F(m \otimes k),$$

so $\tilde{\hat{F}} = F$. And similarly

$$\hat{\tilde{G}}(m)(k) = \tilde{G}(m \otimes k) = G(m)(k),$$

so $\hat{\tilde{G}} = G$.

Naturality is an easy tedious computation.

- (4) We define a natural isomorphism $\text{hom}_S(S, N) \rightarrow N_R$ by mapping $f \in \text{hom}_S(S, N)$ to $f(1)$; denote this by ev_1 , so $\text{ev}_1(f) = f(1)$. This is clearly additive and is R -linear since

$$\text{ev}_1(rf) = (rf)(1) = f(r) = rf(1) = r\text{ev}_1(f).$$

Note that $f(r) = rf(1)$ since f is an S -module morphism and thus an R -module morphism via the map $R \rightarrow S$.

Furthermore it is injective since if $f(1) = 0$ then $f(s) = sf(1) = 0$ for all $s \in S$, so $f = 0$. And finally it is surjective: for $n \in N$, define $f: S \rightarrow N$ by $f(s) = sn$. This is clearly S -linear and $f(1) = n$. Thus ev_1 is an isomorphism, easily seen to be natural.

Thus we have an adjunction

$$\bullet^S = \bullet \otimes_R S \vdash \text{hom}_S(S, \bullet) \cong \bullet_R$$

as desired.

5.6 Problem

Let R be a ring, M, N be fg R -modules.

- (1) Assume R is local with maximal ideal $\mathfrak{m}$.
 - (i) Show that $M \cong 0$ iff $M/\mathfrak{m}M \cong 0$.
 - (ii) Show that $M \otimes_R N \cong 0$ iff $M \cong 0$ or $N \cong 0$.
- (2) Let $f: M \rightarrow M$ be surjective, show that it is an isomorphism.

(1)

- (i) If $M \cong 0$ then $M/\mathfrak{m}M \cong 0$ clearly. Conversely, if $M/\mathfrak{m}M \cong 0$ then $\mathfrak{m}M = M$, and by Nakayama's lemma this means $M \cong 0$. (Nakayama's lemma being $\text{Jac}(R)M = M$ implying $M \cong 0$, in this case $\text{Jac}(R) = \mathfrak{m}$ since it is local.)
- (ii) Note that in general for an ideal $I \leq R$,

$$\frac{M}{IM} \otimes_{R/I} \frac{N}{IN} \cong \left(M \otimes_R \frac{R}{I} \right) \otimes_{R/I} \left(N \otimes_R \frac{R}{I} \right) \cong M \otimes_R N \otimes_R \frac{R}{I} \cong \frac{M \otimes_R N}{I(M \otimes_R N)}.$$

Thus if $M \otimes_R N \cong 0$ then

$$\frac{M}{\mathfrak{m}M} \otimes_{R/\mathfrak{m}} \frac{N}{\mathfrak{m}N} \cong \frac{M \otimes_R N}{\mathfrak{m}(M \otimes_R N)} \cong 0.$$

But $R/\mathfrak{m}$ is a field, and over a field $\dim_K(V \otimes_K W) = (\dim_K V)(\dim_K W)$, so the tensor is zero iff $V \cong 0$ or $W \cong 0$. So $M/\mathfrak{m}M$ or $N/\mathfrak{m}N$ is zero, and by the above point, either $M \cong 0$ or $N \cong 0$.

Conversely the tensor of any module with the zero module is zero.

- (2) We view M as an $R[x]$ module by $x \cdot m = f(m)$ for all $m \in M$. This is well-defined since the R -module structure on M endows $\text{end}(M)$ with an R -module structure $R \rightarrow \text{end}(M)$. Then $R[x]$ is the free R -algebra in a single indeterminant, and so there is a unique morphism $R[x] \rightarrow \text{end}(M)$ mapping x to f commuting with the maps $R \rightarrow R[x], \text{end}(M)$; that is to say, this extends M 's R -module structure to an $R[x]$ -module structure uniquely.

The $R[x]$ -module structure must be given by $p(x) \cdot m = p(f)(m)$ for all $m \in M$ (where $p(f)$ denotes substituting f for x in the polynomial).

In any case, the assumption translates to the fact that $x \cdot M = M$ and thus $(x) \cdot M = M$, where $(x) \leq R[x]$ is the ideal generated by x . Since M is fg as an R -module, it is fg as an $R[x]$ -module, and thus we can apply Nakayama's lemma: there is a $p \in (x)$ such that $p \cdot m = m$ for all $m \in M$. $p \in (x)$ means that $p = x \cdot q$ for some polynomial $q \in R[x]$, and then

$$m = p \cdot m = f \circ q(f)(m) = q(f) \circ f(m).$$

Thus $q(f)$ is f 's inverse, so it is an isomorphism as desired.

5.7 Problem

- (1) Find a natural bijection between the closed points of $|\mathrm{Spec}(\mathbb{Q}(x) \otimes_{\mathbb{Q}} \mathbb{C})|$ and the set of transcendental complex numbers.
- (2) Consider the Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ which we identify with $\lim_K \mathrm{Gal}(K/\mathbb{Q})$ for $K/\mathbb{Q}$ finite Galois. Equipping $\mathrm{Gal}(K/\mathbb{Q})$ with the discrete topology induces a topology on $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ called the **Krull topology**. Prove that $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ and $\mathrm{Spec}(\overline{\mathbb{Q}} \otimes_{\mathbb{Q}} \overline{\mathbb{Q}})$ are homeomorphic.

- (1) In general if A, B are R -algebras and $S \subseteq A$ is multiplicatively closed,

$$S^{-1}A \otimes_R B \cong (S^{-1}A \otimes_A A) \otimes_R B \cong S^{-1}A \otimes_A (A \otimes_R B).$$

Since localization commutes with tensoring, this is isomorphic to

$$S^{-1}A \otimes_R B \cong S^{-1}(A \otimes_A (A \otimes_R B)) \cong S^{-1}(A \otimes_R B).$$

In our case, $\mathbb{Q}(x)$ is the localization of $\mathbb{Q}[x]$ with respect to $S = \mathbb{Q}[x] - 0$, and so

$$\mathbb{Q}(x) \otimes_{\mathbb{Q}} \mathbb{C} \cong S^{-1}(\mathbb{Q}[x] \otimes_{\mathbb{Q}} \mathbb{C}) \cong S^{-1}\mathbb{C}[x],$$

where the final isomorphism is due to an above point. Thus

$$\mathbb{Q}(x) \otimes_{\mathbb{Q}} \mathbb{C} \cong \left\{ \frac{f(x)}{p(x)} \mid f(x) \in \mathbb{C}[x], 0 \neq p(x) \in \mathbb{Q}[x] \right\}.$$

Recall that maximal ideals of $S^{-1}A$ are all of the form $S^{-1}\mathfrak{m}$ where $\mathfrak{m} \leq A$ is an ideal maximal in the property of not intersecting S . Since maximal ideals are prime, and prime ideals of $S^{-1}A$ are of the form $S^{-1}\mathfrak{p}$ for $\mathfrak{p} \leq A$ prime, $\mathfrak{m}$ is prime. In our case, prime ideals of $\mathbb{C}[x]$ are maximal and of the form $(x - \lambda)$ for $\lambda \in \mathbb{C}$. Thus maximal ideals of $\mathbb{Q}(x) \otimes_{\mathbb{Q}} \mathbb{C}$ are the localization of maximal ideals $(x - \lambda)$ which don't intersect $\mathbb{Q}[x] - 0$, equivalently $(x - \lambda) \cap \mathbb{Q}[x] = 0$.

An ideal $(x - \lambda)$ intersects $\mathbb{Q}[x]$ trivially iff λ is transcendental: if $(x - \lambda)p(x) \in \mathbb{Q}[x]$ then this gives a rational polynomial with root λ , so it is algebraic. Conversely if λ is algebraic then $q(\lambda) = 0$ for some $q \in \mathbb{Q}[x]$ and thus $q(x) = (x - \lambda)p(x) \in (x - \lambda) \cap \mathbb{Q}[x]$ for some $p \in \mathbb{C}[x]$.

Thus the bijection in question is given by

$$\lambda \leftrightarrow S^{-1}(x - \lambda).$$

- (2) We have that $\overline{\mathbb{Q}} = \mathrm{colim}_K K$ as K goes through finite Galois extensions of $\mathbb{Q}$ (this is functorial due to restriction). This is a direct colimit, as two finite Galois extensions are contained in their compositum. Tensoring is left-adjoint and thus commutes with colimits:

$$\overline{\mathbb{Q}} \otimes_{\mathbb{Q}} \overline{\mathbb{Q}} \cong \mathrm{colim}_{K, K'} K \otimes_{\mathbb{Q}} K'.$$

Notice that the the product of two directed sets is directed, so this is again a direct colimit. We can restrict direct colimits to cofinal subsets, and the diagonal is cofinal, so this is isomorphic to

$$\overline{\mathbb{Q}} \otimes_{\mathbb{Q}} \overline{\mathbb{Q}} \cong \mathrm{colim}_K K \otimes_{\mathbb{Q}} K.$$

Now, Spec is an antiequivalence and in particular preserves colimits (since it is contravariant it turns colimits into limits). Thus

$$\mathrm{Spec}(\overline{\mathbb{Q}} \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}) \cong \lim_K \mathrm{Spec}(K \otimes_{\mathbb{Q}} K).$$

By the question above, $K \otimes_{\mathbb{Q}} K$ is isomorphic to $\prod_{\text{Gal}(K/\mathbb{Q})} K \cong K^n$ where $n = |\text{Gal}(K/\mathbb{Q})|$.

Since K is a field and thus Noetherian, K^n is Noetherian as well (the property is preserved under finite direct sums). If $v, u \in K^n$ are vectors, then we can normalize them so that all of their components are zero or one and this does not change the ideal they generate (v, u) . If we let w be the vector obtained by taking their “union” ($w_i = 1$ iff v_i or $u_i = 1$), then we see that $(v, u) = (w)$. Thus inductively we see K^n is a PID whose ideals are generated by vectors whose components are all 0 or 1. Notice that $(v) \subseteq (u)$ iff the set of indices where $u_i = 1$ contains the set of indices where $v_i = 1$. Thus the maximal ideals of K^n are those generated by vectors v which contain only a single zero. These are also the prime ideals: if $v_i = v_j = 0$ for $i \neq j$, then let u^i be v but whose i th component is 1, then $u^i u^j = v$ but neither are in (v) .

Thus $|\text{Spec}(K^n)| = \text{Spec}(K^n)$ has size n . Since it is finite and all points are closed, $\text{Spec}(K^n)$ is homeomorphic to $[n]$, the discrete space of size n . In particular

$$\text{Spec}(K \otimes_{\mathbb{Q}} K) \cong \text{Spec}\left(\prod_{\text{Gal}(K/\mathbb{Q})} K\right) \cong \text{Gal}(K/\mathbb{Q}),$$

since $\text{Gal}(K/\mathbb{Q})$ was given the discrete topology.

And thus

$$\text{Spec}(\overline{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{Q}) \cong \lim_K \text{Gal}(K/\mathbb{Q}) \cong \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$$

as desired.

5.8 Problem

Consider the **combinatorial Nullstellensatz**:

Theorem (Combinatorial Nullstellensatz)

Let k be a field, $f \in k[x_1, \dots, x_n]$. Suppose $x^{t_1} \dots x^{t_n}$ is a monomial of maximal degree appearing in f (that is, $\sum_i t_i$ is maximal). Let $S_1, \dots, S_n \subseteq k$ be subsets such that $|S_i| > t_i$ for all i . Then there are $s_i \in S_i$ such that $f(s_1, \dots, s_n) \neq 0$.

Use it to prove the following:

Theorem (Cauchy-Davenport)

Let p be prime, and $A, B \subseteq \mathbb{F}_p$ nonempty such that $|A| + |B| - 1 \leq p$. Then $|A + B| \geq |A| + |B| - 1$.

Suppose this were not the case, so $|A + B| \leq |A| + |B| - 2 < p$. Let us extend $A + B$ to a subset $C \subseteq \mathbb{F}_p$ of size $n = |A| + |B| - 2$. Define $f \in \mathbb{F}_p[x, y]$ by

$$f(x, y) = \prod_{c \in C} (x + y - c).$$

The maximal degree of f is $|C| = n$, and by the binomial theorem, the coefficient of $x^k y^{n-k}$ in f is $\binom{n}{k}$, which is nonzero since $n < p$. In particular for $k = |A| - 1$ we have $n - k = |B| - 1$, so $|A| > k, |B| > n - k$ and thus by the combinatorial Nullstellensatz there are $a \in A, b \in B$ such that $f(a, b) \neq 0$. But then $a + b \notin C$ and in particular $a + b \notin A + B$, a contradiction.